CODE EXPLANATIONS

We first start by importing all packages in the Jupyter/Python environments. We then import label mapping functions that will track the images by splitting them up into a series of horizontal and vertical vectors.

Object Detection Results

The third part of the code specifies the image is importing functions along with the commands for downloading neural network schemas to be used for training the datasets. Likewise, the format for the input and output are entirely influenced based on the sort of model used which is seen in the next code snippet.

This part of the code tells the system the required model that should be used along with the frozen inference graph that will be generated by TensorFlow to compare how well the images are being learnt by the system.

The DOWNLOAD_BASE, MODEL_NAME, MODEL_FILE all refer to the model that will be scanning the images and using different hidden layers to produce a benchmarked classification system on how the object can be detected from the images.

In the meantime, once the code is running, the system generates labels for all the various kinds of objects that may be available in the images: the number of unique objects available, the more time the model will take to distinguish each of them and then associate them with certain attributes.

The images will thus be converted into an array of numerical attributes depending on the location, dimensions, size, depth and angle. It's important to understand that object colour has little effect on the way the system learns and labels the different items as all images are converted into a monochromatic colour scheme to better aid the network.

numPy is the extension that plays an important role in this. Moving on, the next snippet creates a naming convention for all the items

The next snippet of code creates a path to the images for testing them and builds a naming convention so that they can be referenced by the network easily.

The following snippet is used to create an inference for a single image and stores the location of the unique object as an array of numbers denoting position and size. Additionally, this alerts the system to create boxes around